

Heartbeat-evoked potentials vary with clinical diagnosis, sleep stage and age

Nir Cafri^{1,3}, Felix Benninger^{2,3}, Pablo Blinder^{1,2}

¹Tel Aviv University; ²Sagol School of Neuroscience; ³Rabin Medical Center, Israel. Correspondence: nircafri@mail.tau.ac.il

Abstract

Heartbeat-evoked potentials (HEPs) provide a non-invasive measure of cortical processing of cardiac afferent signals, but their clinical specificity must be separated from normal sleep- and age-related variation. We analysed R-peak-locked EEG in a large clinical polysomnography resource spanning 2,931 patients in the stage-comparison cohort and 19 standard scalp electrodes. Diagnosis groups differed within matched sleep stages, while mixed-effects modelling adjusted these effects for age, sex, heart rate and cardiac-field contamination. Within patients, significant stage contrasts followed a graded REM/light-sleep-to-N3 pattern. Age effects were significant across sleep stages and were spatially structured. Together, these results establish diagnosis, vigilance state and age as separable axes of HEP variability and motivate stage-matched, covariate-aware clinical use.

Introduction

The HEP is a small EEG deflection time-locked to the electrocardiographic R peak. It is interpreted as a marker of cortical interoception and self-referential processing,^{5,9,10} yet it is measured beside a much larger cardiac field artifact and is sensitive to arousal, so clinical studies require direct visualization of the source signals, explicit exclusion of the peri-R interval, and comparison within the same sleep state, following recent methodological guidance on HEP analysis and reporting.⁷ HEP amplitude follows a known sleep-stage gradient² and increases with age in wakefulness,^{3,4} effects we previously reported in preliminary form.⁸ Here we integrate representative physiology, cohort-level waveforms, diagnosis comparisons, within-patient sleep-stage contrasts and age effects in one concise analysis.

Signal-level origin of the HEP

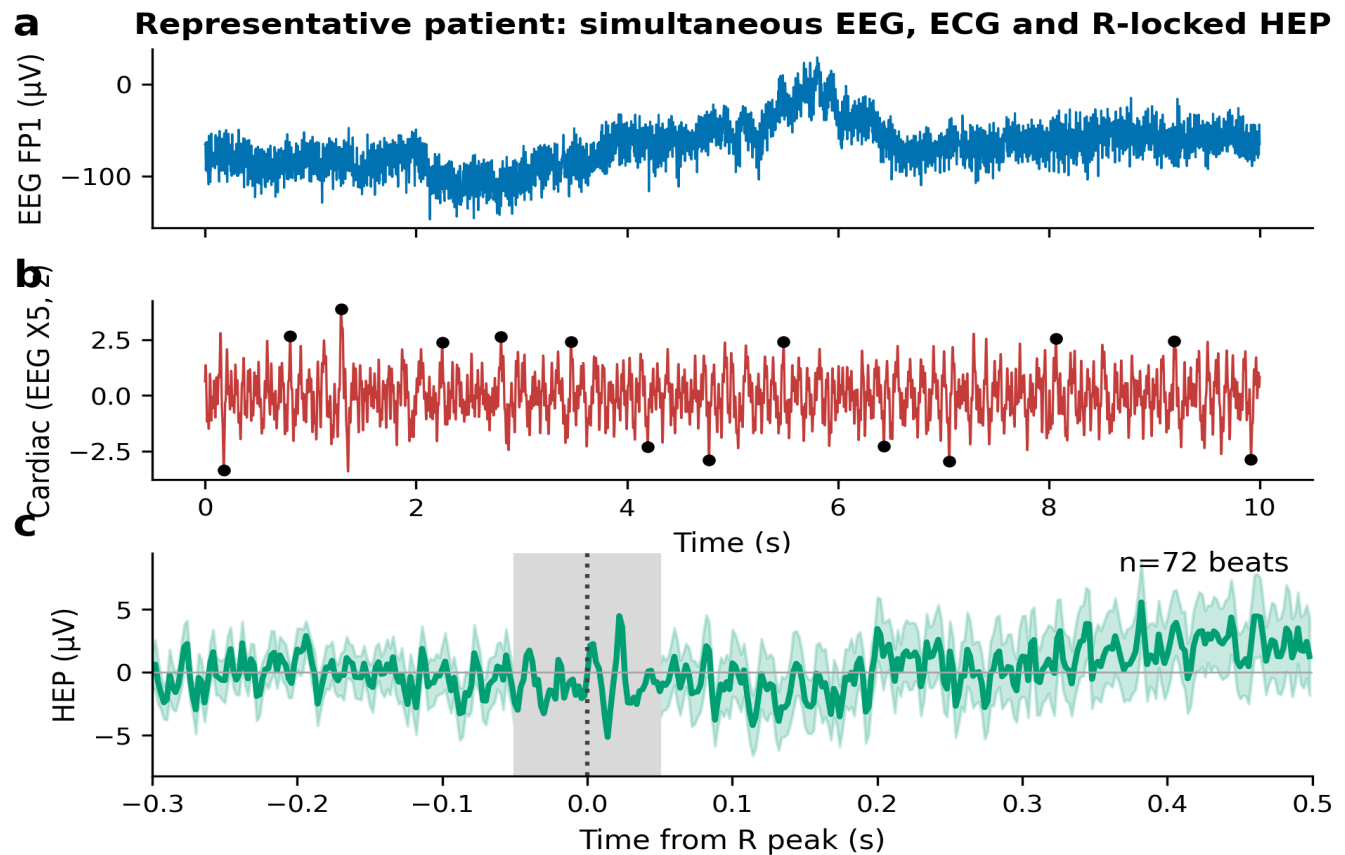


Figure 1 | Representative simultaneous EEG and cardiac activity from an actual patient recording. R peaks (black markers) define epochs; the grey band marks the -50 to +50 ms cardiac-field-artifact interval excluded from inference. The lower trace is the patient-level mean $\pm$ s.e.m.

Cohort HEP and electrode coverage

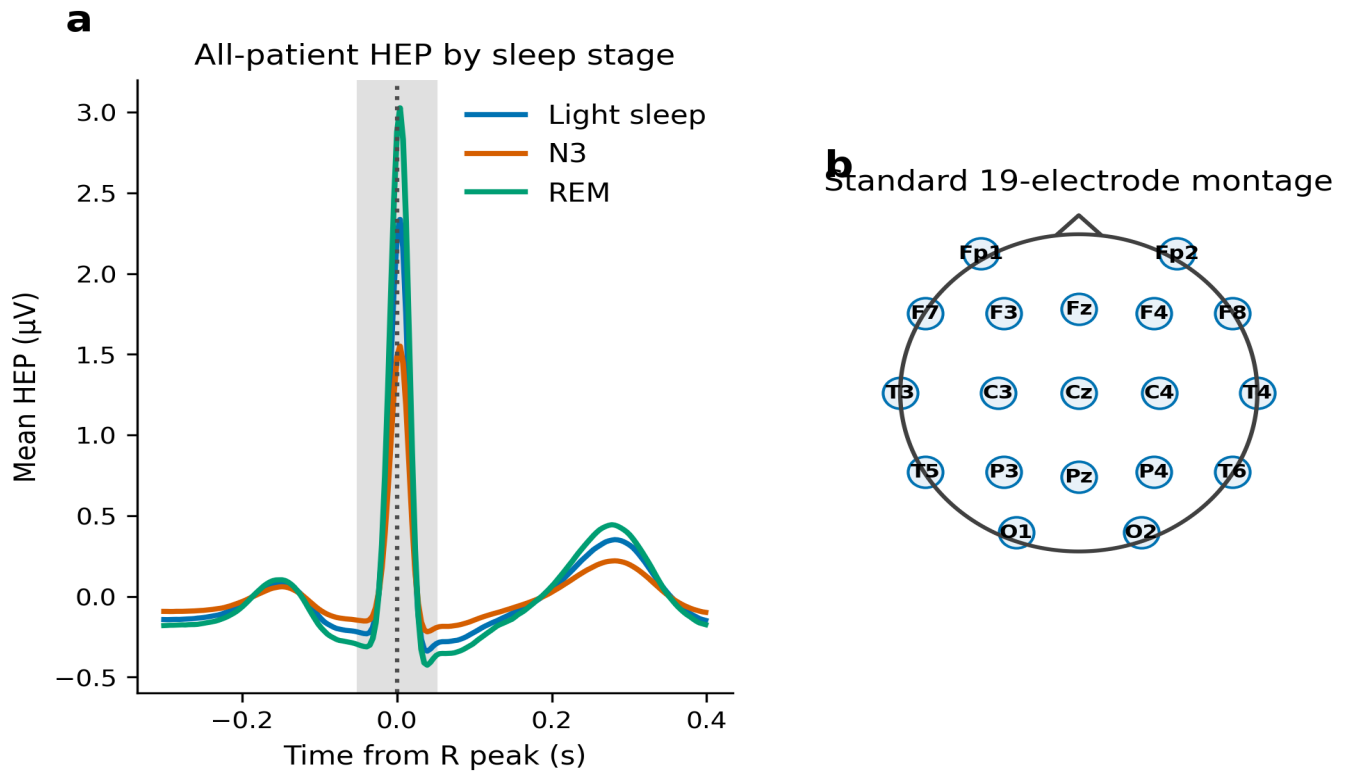


Figure 2 | Stage-resolved all-patient HEP waveforms and the standard 19-electrode 10–20 montage used for topographic inference. The common spatial frame prevents montage differences from being mistaken for biology.

Diagnosis effects within the same sleep stage

Diagnosis comparisons were performed separately within light sleep, N3 and REM. The saved Kruskal–Wallis analyses identified group-level heterogeneity in every stage; the accompanying mixed-effects model retained diagnosis terms while controlling for stage, scalp region, age, sex, heart rate and cardiac-field-artifact amplitude. These analyses support a diagnosis-associated HEP component, but do not imply that the waveform alone is diagnostically specific.

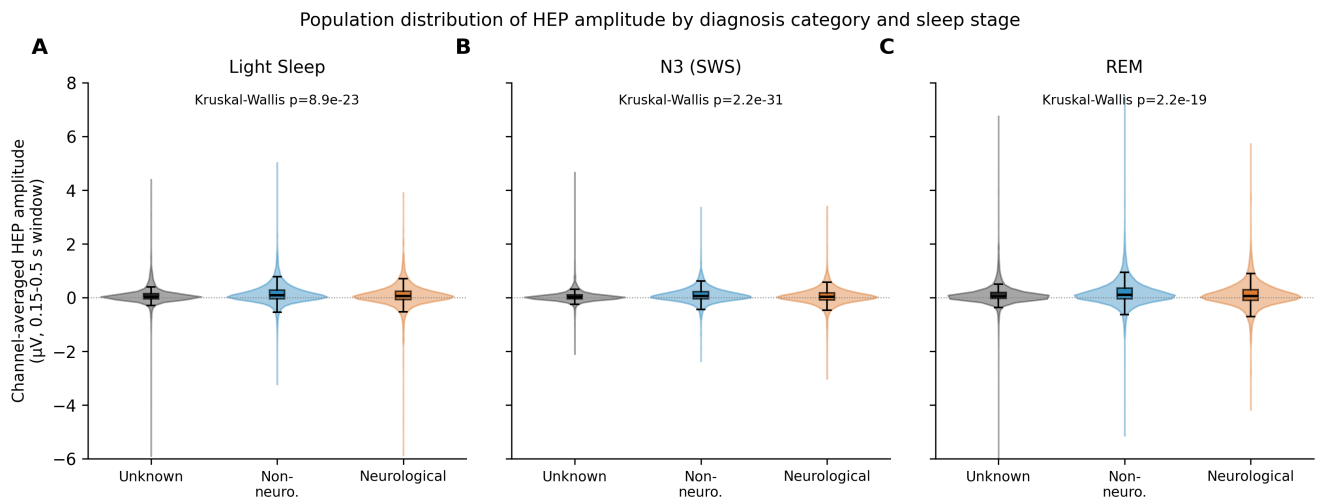


Figure 3 | Stage-matched diagnosis comparison. Channel-averaged HEP amplitude is shown by broad diagnostic class within each sleep stage. Omnibus statistics and sample sizes are reported in the panels; comparisons never pool different vigilance states.

Within-diagnosis sleep-stage effects

Significant within-patient sleep-stage contrasts

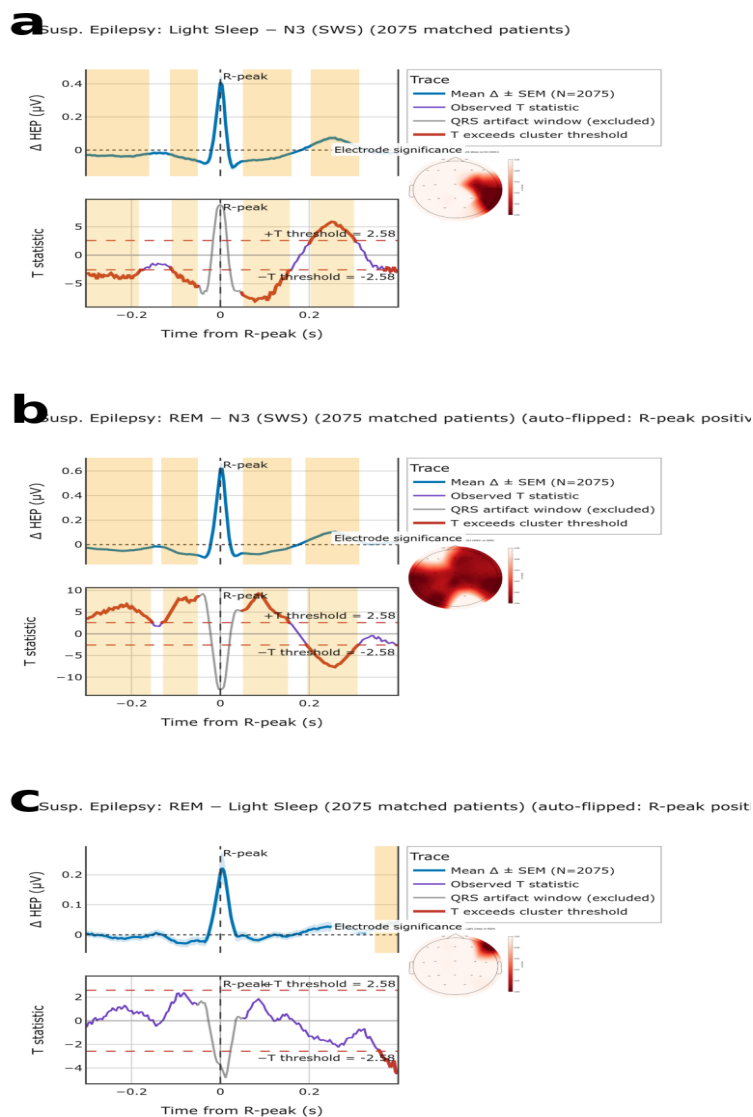


Figure 4 | Only omnibus-significant paired stage contrasts are displayed. Each row combines the mean difference waveform with electrode-wise topography. Red electrodes pass the prespecified cluster threshold; the peri-R grey interval is excluded.

Age effects across sleep stages

Age-associated HEP differences across sleep

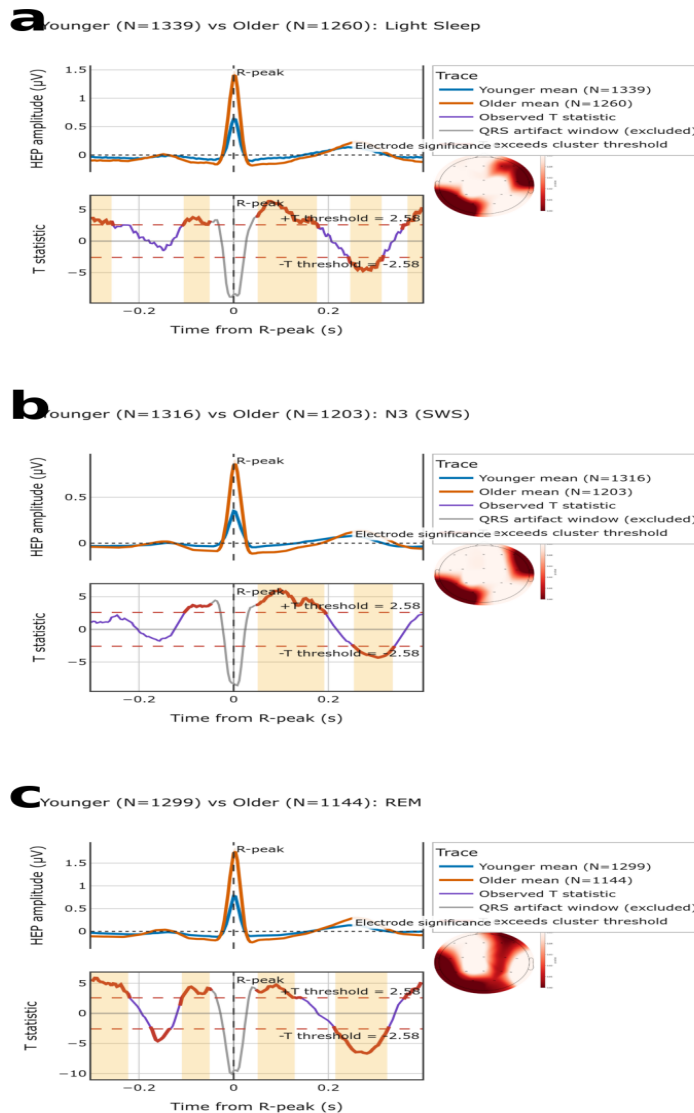


Figure 5 | Older-versus-younger median-split contrasts, shown only for stages meeting the omnibus significance criterion. Waveform and 19-electrode maps demonstrate that ageing is a spatially organized covariate rather than a uniform offset.

Discussion

Three conclusions emerge. First, a measurable post-R HEP remains after the artifact-dominated interval and can be visualized at the individual and cohort levels. Second, clinical groups differ even when comparisons are restricted to the same sleep stage and adjusted for major physiologic covariates. Third, sleep stage and age produce large, topographically coherent shifts that can confound unstratified disease comparisons. The strongest design for future biomarker studies is therefore within-stage, montage-matched and age-adjusted, with patient-level replication and independent validation.

Methods summary

This multi-hospital design parallels prior multicentre epilepsy imaging work.¹

EEG was epoched around detected R peaks and baseline-corrected. The -50 to +50 ms interval was treated as cardiac-field artifact and excluded from statistical interpretation. Diagnosis was tested within sleep stage using non-parametric omnibus comparisons and in a patient-random-intercept mixed model with stage, region, age, sex, heart rate and artifact covariates. Paired sleep-stage differences and independent age contrasts were evaluated electrode-wise with cluster-mass permutation testing, following the nonparametric cluster-based framework for EEG/MEG data,⁶ using 200 permutations and cluster $\alpha=0.01$, with only significant panels retained in Figures 4–5. Values and figures were read from the frozen analysis result files in Paper1; no statistics were recomputed or invented. Aggregate outputs and the

builder are in Paper1; patient-level data remain governed by source-cohort access conditions.

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